SG No. 210 O^4 $F4_132$ [cubic]

* plus set: $+[0,\,0,\,0]$, $+\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$
* Wyckoff site: `8a`, site symmetry: `23.`

Table 1: Wyckoff bond: `24a@8a`, bond symmetry: `222..`

No.	vector	center	mapping
1	$[X,\,0,\,0]$	$[0,\,0,\,0]$	`[1,-2,-3,4]`
2	$[0,\,X,\,0]$	$[0,\,0,\,0]$	`[5,-6,-7,8]`
3	$[0,\,0,\,X]$	$[0,\,0,\,0]$	`[9,-10,-11,12]`
4	$[0,\,X,\,0]$	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[13,-14,-15,16]`
5	$[X,\,0,\,0]$	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[17,-18,-19,20]`
6	$[0,\,0,\,-X]$	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[21,-22,-23,24]`

Table 2: Wyckoff bond: `32b@8a`, bond symmetry: `.3.`

No.	vector	center	mapping
1	$[X,\,X,\,X]$	$[0,\,0,\,0]$	`[1,5,9]`
2	$[-X,\,-X,\,X]$	$[0,\,0,\,0]$	`[2,7,12]`
3	$[-X,\,X,\,-X]$	$[0,\,0,\,0]$	`[3,8,10]`
4	$[X,\,-X,\,-X]$	$[0,\,0,\,0]$	`[4,6,11]`
5	$[X,\,X,\,-X]$	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[13,17,21]`
6	$[-X,\,-X,\,-X]$	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[14,19,24]`
7	$[X,\,-X,\,X]$	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[15,20,22]`
8	$[-X,\,X,\,X]$	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[16,18,23]`

Table 3: Wyckoff bond: `48c@8a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\,Y,\,0]$	$[0,\,0,\,0]$	`[1,-2]`
2	$[-X,\,Y,\,0]$	$[0,\,0,\,0]$	`[3,-4]`
3	$[0,\,X,\,Y]$	$[0,\,0,\,0]$	`[5,-6]`
4	$[0,\,-X,\,Y]$	$[0,\,0,\,0]$	`[7,-8]`
5	$[Y,\,0,\,X]$	$[0,\,0,\,0]$	`[9,-10]`
6	$[Y,\,0,\,-X]$	$[0,\,0,\,0]$	`[11,-12]`
7	$[Y,\,X,\,0]$	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[13,-14]`
8	$[Y,\,-X,\,0]$	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[15,-16]`
9	$[X,\,0,\,-Y]$	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[17,-18]`
10	$[-X,\,0,\,-Y]$	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[19,-20]`
11	$[0,\,Y,\,-X]$	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[21,-22]`
12	$[0,\,Y,\,X]$	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[23,-24]`

Table 4: Wyckoff bond: `96d@8a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\,Y,\,Z]$	$[0,\,0,\,0]$	`[1]`

continued ...

Table 4

No.	vector	center	mapping
2	$[-X, -Y, Z]$	$[0, 0, 0]$	`[2]`
3	$[-X, Y, -Z]$	$[0, 0, 0]$	`[3]`
4	$[X, -Y, -Z]$	$[0, 0, 0]$	`[4]`
5	$[Z, X, Y]$	$[0, 0, 0]$	`[5]`
6	$[Z, -X, -Y]$	$[0, 0, 0]$	`[6]`
7	$[-Z, -X, Y]$	$[0, 0, 0]$	`[7]`
8	$[-Z, X, -Y]$	$[0, 0, 0]$	`[8]`
9	$[Y, Z, X]$	$[0, 0, 0]$	`[9]`
10	$[-Y, Z, -X]$	$[0, 0, 0]$	`[10]`
11	$[Y, -Z, -X]$	$[0, 0, 0]$	`[11]`
12	$[-Y, -Z, X]$	$[0, 0, 0]$	`[12]`
13	$[Y, X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[13]`
14	$[-Y, -X, -Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[14]`
15	$[Y, -X, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[15]`
16	$[-Y, X, Z]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[16]`
17	$[X, Z, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[17]`
18	$[-X, Z, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[18]`
19	$[-X, -Z, -Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[19]`
20	$[X, -Z, Y]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[20]`
21	$[Z, Y, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[21]`
22	$[Z, -Y, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[22]`
23	$[-Z, Y, X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[23]`
24	$[-Z, -Y, -X]$	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[24]`

* Wyckoff site: `8b`, site symmetry: `23.`

Table 5: Wyckoff bond: `24a@8b`, bond symmetry: `222..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[1,-2,-3,4]`
2	$[0, X, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[5,-6,-7,8]`
3	$[0, 0, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[9,-10,-11,12]`
4	$[0, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[13,-14,-15,16]`
5	$[X, 0, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[17,-18,-19,20]`
6	$[0, 0, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[21,-22,-23,24]`

Table 6: Wyckoff bond: `32b@8b`, bond symmetry: `.3.`

No.	vector	center	mapping
1	$[X, X, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[1,5,9]`
2	$[-X, -X, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[2,7,12]`
3	$[-X, X, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[3,8,10]`
4	$[X, -X, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[4,6,11]`
5	$[X, X, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[13,17,21]`
6	$[-X, -X, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[14,19,24]`
7	$[X, -X, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[15,20,22]`

continued ...

Table 6

No.	vector	center	mapping
8	$[-X, X, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[16,18,23]`

Table 7: Wyckoff bond: `48c@8b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[1,-2]`
2	$[-X, Y, 0]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[3,-4]`
3	$[0, X, Y]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[5,-6]`
4	$[0, -X, Y]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[7,-8]`
5	$[Y, 0, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[9,-10]`
6	$[Y, 0, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[11,-12]`
7	$[Y, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[13,-14]`
8	$[Y, -X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[15,-16]`
9	$[X, 0, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[17,-18]`
10	$[-X, 0, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[19,-20]`
11	$[0, Y, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[21,-22]`
12	$[0, Y, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[23,-24]`

Table 8: Wyckoff bond: `96d@8b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[2]`
3	$[-X, Y, -Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[3]`
4	$[X, -Y, -Z]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[4]`
5	$[Z, X, Y]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[5]`
6	$[Z, -X, -Y]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[6]`
7	$[-Z, -X, Y]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[7]`
8	$[-Z, X, -Y]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[8]`
9	$[Y, Z, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[9]`
10	$[-Y, Z, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[10]`
11	$[Y, -Z, -X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[11]`
12	$[-Y, -Z, X]$	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[12]`
13	$[Y, X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[13]`
14	$[-Y, -X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[14]`
15	$[Y, -X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[15]`
16	$[-Y, X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[16]`
17	$[X, Z, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[17]`
18	$[-X, Z, Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[18]`
19	$[-X, -Z, -Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[19]`
20	$[X, -Z, Y]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[20]`
21	$[Z, Y, -X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[21]`
22	$[Z, -Y, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[22]`
23	$[-Z, Y, X]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[23]`

continued ...

Table 8

No.	vector	center	mapping
24	$[-Z, -Y, -X]$	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[24]`

* Wyckoff site: `16c`, site symmetry: `.32`

Table 9: Wyckoff bond: `16a@16c`, bond symmetry: `.32`

No.	vector	center	mapping
1	$[X, X, X]$	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right]$	`[1,5,9,-14,-19,-24]`
2	$[-X, -X, X]$	$\left[\frac{7}{8}, \frac{3}{8}, \frac{5}{8}\right]$	`[2,7,12,-13,-17,-21]`
3	$[-X, X, -X]$	$\left[\frac{3}{8}, \frac{5}{8}, \frac{7}{8}\right]$	`[3,8,10,-15,-20,-22]`
4	$[X, -X, -X]$	$\left[\frac{5}{8}, \frac{7}{8}, \frac{3}{8}\right]$	`[4,6,11,-16,-18,-23]`

Table 10: Wyckoff bond: `48b@16c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Y]$	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right]$	`[1,-14]`
2	$[-X, -X, Y]$	$\left[\frac{7}{8}, \frac{3}{8}, \frac{5}{8}\right]$	`[2,-13]`
3	$[-X, X, -Y]$	$\left[\frac{3}{8}, \frac{5}{8}, \frac{7}{8}\right]$	`[3,-15]`
4	$[X, -X, -Y]$	$\left[\frac{5}{8}, \frac{7}{8}, \frac{3}{8}\right]$	`[4,-16]`
5	$[Y, X, X]$	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right]$	`[5,-24]`
6	$[Y, -X, -X]$	$\left[\frac{5}{8}, \frac{7}{8}, \frac{3}{8}\right]$	`[6,-23]`
7	$[-Y, -X, X]$	$\left[\frac{7}{8}, \frac{3}{8}, \frac{5}{8}\right]$	`[7,-21]`
8	$[-Y, X, -X]$	$\left[\frac{3}{8}, \frac{5}{8}, \frac{7}{8}\right]$	`[8,-22]`
9	$[X, Y, X]$	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right]$	`[9,-19]`
10	$[-X, Y, -X]$	$\left[\frac{3}{8}, \frac{5}{8}, \frac{7}{8}\right]$	`[10,-20]`
11	$[X, -Y, -X]$	$\left[\frac{5}{8}, \frac{7}{8}, \frac{3}{8}\right]$	`[11,-18]`
12	$[-X, -Y, X]$	$\left[\frac{7}{8}, \frac{3}{8}, \frac{5}{8}\right]$	`[12,-17]`

Table 11: Wyckoff bond: `48c@16c`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, 0]$	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right]$	`[1,14]`
2	$[-X, X, 0]$	$\left[\frac{7}{8}, \frac{3}{8}, \frac{5}{8}\right]$	`[2,13]`
3	$[-X, -X, 0]$	$\left[\frac{3}{8}, \frac{5}{8}, \frac{7}{8}\right]$	`[3,15]`
4	$[X, X, 0]$	$\left[\frac{5}{8}, \frac{7}{8}, \frac{3}{8}\right]$	`[4,16]`
5	$[0, X, -X]$	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right]$	`[5,24]`
6	$[0, -X, X]$	$\left[\frac{5}{8}, \frac{7}{8}, \frac{3}{8}\right]$	`[6,23]`
7	$[0, -X, -X]$	$\left[\frac{7}{8}, \frac{3}{8}, \frac{5}{8}\right]$	`[7,21]`
8	$[0, X, X]$	$\left[\frac{3}{8}, \frac{5}{8}, \frac{7}{8}\right]$	`[8,22]`
9	$[-X, 0, X]$	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right]$	`[9,19]`
10	$[X, 0, -X]$	$\left[\frac{3}{8}, \frac{5}{8}, \frac{7}{8}\right]$	`[10,20]`
11	$[-X, 0, -X]$	$\left[\frac{5}{8}, \frac{7}{8}, \frac{3}{8}\right]$	`[11,18]`
12	$[X, 0, X]$	$\left[\frac{7}{8}, \frac{3}{8}, \frac{5}{8}\right]$	`[12,17]`

Table 12: Wyckoff bond: `96d@16c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{7}{8}, \frac{3}{8}, \frac{5}{8}]$	`[2]`
3	$[-X, Y, -Z]$	$[\frac{3}{8}, \frac{5}{8}, \frac{7}{8}]$	`[3]`
4	$[X, -Y, -Z]$	$[\frac{5}{8}, \frac{7}{8}, \frac{3}{8}]$	`[4]`
5	$[Z, X, Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[5]`
6	$[Z, -X, -Y]$	$[\frac{5}{8}, \frac{7}{8}, \frac{3}{8}]$	`[6]`
7	$[-Z, -X, Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{5}{8}]$	`[7]`
8	$[-Z, X, -Y]$	$[\frac{3}{8}, \frac{5}{8}, \frac{7}{8}]$	`[8]`
9	$[Y, Z, X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[9]`
10	$[-Y, Z, -X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{7}{8}]$	`[10]`
11	$[Y, -Z, -X]$	$[\frac{5}{8}, \frac{7}{8}, \frac{3}{8}]$	`[11]`
12	$[-Y, -Z, X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{5}{8}]$	`[12]`
13	$[Y, X, -Z]$	$[\frac{7}{8}, \frac{3}{8}, \frac{5}{8}]$	`[13]`
14	$[-Y, -X, -Z]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[14]`
15	$[Y, -X, Z]$	$[\frac{3}{8}, \frac{5}{8}, \frac{7}{8}]$	`[15]`
16	$[-Y, X, Z]$	$[\frac{5}{8}, \frac{7}{8}, \frac{3}{8}]$	`[16]`
17	$[X, Z, -Y]$	$[\frac{7}{8}, \frac{3}{8}, \frac{5}{8}]$	`[17]`
18	$[-X, Z, Y]$	$[\frac{5}{8}, \frac{7}{8}, \frac{3}{8}]$	`[18]`
19	$[-X, -Z, -Y]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[19]`
20	$[X, -Z, Y]$	$[\frac{3}{8}, \frac{5}{8}, \frac{7}{8}]$	`[20]`
21	$[Z, Y, -X]$	$[\frac{7}{8}, \frac{3}{8}, \frac{5}{8}]$	`[21]`
22	$[Z, -Y, X]$	$[\frac{3}{8}, \frac{5}{8}, \frac{7}{8}]$	`[22]`
23	$[-Z, Y, X]$	$[\frac{5}{8}, \frac{7}{8}, \frac{3}{8}]$	`[23]`
24	$[-Z, -Y, -X]$	$[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}]$	`[24]`

* Wyckoff site: `16d`, site symmetry: `.32`

Table 13: Wyckoff bond: `16a@16d`, bond symmetry: `.32`

No.	vector	center	mapping
1	$[X, X, X]$	$[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}]$	`[1,5,9,-14,-19,-24]`
2	$[-X, -X, X]$	$[\frac{3}{8}, \frac{7}{8}, \frac{1}{8}]$	`[2,7,12,-13,-17,-21]`
3	$[-X, X, -X]$	$[\frac{7}{8}, \frac{1}{8}, \frac{3}{8}]$	`[3,8,10,-15,-20,-22]`
4	$[X, -X, -X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{7}{8}]$	`[4,6,11,-16,-18,-23]`

Table 14: Wyckoff bond: `48b@16d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Y]$	$[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}]$	`[1,-14]`
2	$[-X, -X, Y]$	$[\frac{3}{8}, \frac{7}{8}, \frac{1}{8}]$	`[2,-13]`
3	$[-X, X, -Y]$	$[\frac{7}{8}, \frac{1}{8}, \frac{3}{8}]$	`[3,-15]`
4	$[X, -X, -Y]$	$[\frac{1}{8}, \frac{3}{8}, \frac{7}{8}]$	`[4,-16]`
5	$[Y, X, X]$	$[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}]$	`[5,-24]`
6	$[Y, -X, -X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{7}{8}]$	`[6,-23]`
7	$[-Y, -X, X]$	$[\frac{3}{8}, \frac{7}{8}, \frac{1}{8}]$	`[7,-21]`

continued ...

Table 14

No.	vector	center	mapping
8	$[-Y, X, -X]$	$[\frac{7}{8}, \frac{1}{8}, \frac{3}{8}]$	`[8,-22]`
9	$[X, Y, X]$	$[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}]$	`[9,-19]`
10	$[-X, Y, -X]$	$[\frac{7}{8}, \frac{1}{8}, \frac{3}{8}]$	`[10,-20]`
11	$[X, -Y, -X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{7}{8}]$	`[11,-18]`
12	$[-X, -Y, X]$	$[\frac{3}{8}, \frac{7}{8}, \frac{1}{8}]$	`[12,-17]`

Table 15: Wyckoff bond: `48c@16d`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, 0]$	$[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}]$	`[1,14]`
2	$[-X, X, 0]$	$[\frac{3}{8}, \frac{7}{8}, \frac{1}{8}]$	`[2,13]`
3	$[-X, -X, 0]$	$[\frac{7}{8}, \frac{1}{8}, \frac{3}{8}]$	`[3,15]`
4	$[X, X, 0]$	$[\frac{1}{8}, \frac{3}{8}, \frac{7}{8}]$	`[4,16]`
5	$[0, X, -X]$	$[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}]$	`[5,24]`
6	$[0, -X, X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{7}{8}]$	`[6,23]`
7	$[0, -X, -X]$	$[\frac{3}{8}, \frac{7}{8}, \frac{1}{8}]$	`[7,21]`
8	$[0, X, X]$	$[\frac{7}{8}, \frac{1}{8}, \frac{3}{8}]$	`[8,22]`
9	$[-X, 0, X]$	$[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}]$	`[9,19]`
10	$[X, 0, -X]$	$[\frac{7}{8}, \frac{1}{8}, \frac{3}{8}]$	`[10,20]`
11	$[-X, 0, -X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{7}{8}]$	`[11,18]`
12	$[X, 0, X]$	$[\frac{3}{8}, \frac{7}{8}, \frac{1}{8}]$	`[12,17]`

Table 16: Wyckoff bond: `96d@16d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{3}{8}, \frac{7}{8}, \frac{1}{8}]$	`[2]`
3	$[-X, Y, -Z]$	$[\frac{7}{8}, \frac{1}{8}, \frac{3}{8}]$	`[3]`
4	$[X, -Y, -Z]$	$[\frac{1}{8}, \frac{3}{8}, \frac{7}{8}]$	`[4]`
5	$[Z, X, Y]$	$[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}]$	`[5]`
6	$[Z, -X, -Y]$	$[\frac{1}{8}, \frac{3}{8}, \frac{7}{8}]$	`[6]`
7	$[-Z, -X, Y]$	$[\frac{3}{8}, \frac{7}{8}, \frac{1}{8}]$	`[7]`
8	$[-Z, X, -Y]$	$[\frac{7}{8}, \frac{1}{8}, \frac{3}{8}]$	`[8]`
9	$[Y, Z, X]$	$[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}]$	`[9]`
10	$[-Y, Z, -X]$	$[\frac{7}{8}, \frac{1}{8}, \frac{3}{8}]$	`[10]`
11	$[Y, -Z, -X]$	$[\frac{1}{8}, \frac{3}{8}, \frac{7}{8}]$	`[11]`
12	$[-Y, -Z, X]$	$[\frac{3}{8}, \frac{7}{8}, \frac{1}{8}]$	`[12]`
13	$[Y, X, -Z]$	$[\frac{3}{8}, \frac{7}{8}, \frac{1}{8}]$	`[13]`
14	$[-Y, -X, -Z]$	$[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}]$	`[14]`
15	$[Y, -X, Z]$	$[\frac{7}{8}, \frac{1}{8}, \frac{3}{8}]$	`[15]`
16	$[-Y, X, Z]$	$[\frac{1}{8}, \frac{3}{8}, \frac{7}{8}]$	`[16]`
17	$[X, Z, -Y]$	$[\frac{3}{8}, \frac{7}{8}, \frac{1}{8}]$	`[17]`
18	$[-X, Z, Y]$	$[\frac{1}{8}, \frac{3}{8}, \frac{7}{8}]$	`[18]`
19	$[-X, -Z, -Y]$	$[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}]$	`[19]`

continued ...

Table 16

No.	vector	center	mapping
20	$[X, -Z, Y]$	$\left[\frac{7}{8}, \frac{1}{8}, \frac{3}{8}\right]$	[20]
21	$[Z, Y, -X]$	$\left[\frac{3}{8}, \frac{7}{8}, \frac{1}{8}\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{7}{8}, \frac{1}{8}, \frac{3}{8}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{8}, \frac{3}{8}, \frac{7}{8}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}\right]$	[24]

* Wyckoff site: 32e, site symmetry: .3.

Table 17: Wyckoff bond: 32a@32e, bond symmetry: .3.

No.	vector	center	mapping
1	$[X, X, X]$	$[x, x, x]$	[1,5,9]
2	$[-X, -X, X]$	$\left[-x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	[2,7,12]
3	$[-X, X, -X]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, -x\right]$	[3,8,10]
4	$[X, -X, -X]$	$\left[x + \frac{1}{2}, -x, \frac{1}{2} - x\right]$	[4,6,11]
5	$[X, X, -X]$	$\left[x + \frac{3}{4}, x + \frac{1}{4}, \frac{3}{4} - x\right]$	[13,17,21]
6	$[-X, -X, -X]$	$\left[\frac{1}{4} - x, \frac{1}{4} - x, \frac{1}{4} - x\right]$	[14,19,24]
7	$[X, -X, X]$	$\left[x + \frac{1}{4}, \frac{3}{4} - x, x + \frac{3}{4}\right]$	[15,20,22]
8	$[-X, X, X]$	$\left[\frac{3}{4} - x, x + \frac{3}{4}, x + \frac{1}{4}\right]$	[16,18,23]

Table 18: Wyckoff bond: 96b@32e, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, x, x]$	[1]
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, -x\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x + \frac{1}{2}, -x, \frac{1}{2} - x\right]$	[4]
5	$[Z, X, Y]$	$[x, x, x]$	[5]
6	$[Z, -X, -Y]$	$\left[x + \frac{1}{2}, -x, \frac{1}{2} - x\right]$	[6]
7	$[-Z, -X, Y]$	$\left[-x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, -x\right]$	[8]
9	$[Y, Z, X]$	$[x, x, x]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{2} - x, x + \frac{1}{2}, -x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[x + \frac{1}{2}, -x, \frac{1}{2} - x\right]$	[11]
12	$[-Y, -Z, X]$	$\left[-x, \frac{1}{2} - x, x + \frac{1}{2}\right]$	[12]
13	$[Y, X, -Z]$	$\left[x + \frac{3}{4}, x + \frac{1}{4}, \frac{3}{4} - x\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{1}{4} - x, \frac{1}{4} - x, \frac{1}{4} - x\right]$	[14]
15	$[Y, -X, Z]$	$\left[x + \frac{1}{4}, \frac{3}{4} - x, x + \frac{3}{4}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{3}{4} - x, x + \frac{3}{4}, x + \frac{1}{4}\right]$	[16]
17	$[X, Z, -Y]$	$\left[x + \frac{3}{4}, x + \frac{1}{4}, \frac{3}{4} - x\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{3}{4} - x, x + \frac{3}{4}, x + \frac{1}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{4} - x, \frac{1}{4} - x, \frac{1}{4} - x\right]$	[19]
20	$[X, -Z, Y]$	$\left[x + \frac{1}{4}, \frac{3}{4} - x, x + \frac{3}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[x + \frac{3}{4}, x + \frac{1}{4}, \frac{3}{4} - x\right]$	[21]
22	$[Z, -Y, X]$	$\left[x + \frac{1}{4}, \frac{3}{4} - x, x + \frac{3}{4}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{3}{4} - x, x + \frac{3}{4}, x + \frac{1}{4}\right]$	[23]

continued ...

Table 18

No.	vector	center	mapping
24	$[-Z, -Y, -X]$	$\left[\frac{1}{4}-x, \frac{1}{4}-x, \frac{1}{4}-x\right]$	`[24]`

* Wyckoff site: `48f`, site symmetry: `2..`

Table 19: Wyckoff bond: `48a@48f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0, X, Y]$	$[x, 0, 0]$	`[1,-4]`
2	$[0, -X, Y]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[2,-3]`
3	$[Y, 0, X]$	$[0, x, 0]$	`[5,-8]`
4	$[Y, 0, -X]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	`[6,-7]`
5	$[X, Y, 0]$	$[0, 0, x]$	`[9,-12]`
6	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	`[10,-11]`
7	$[X, 0, -Y]$	$\left[\frac{3}{4}, x+\frac{1}{4}, \frac{3}{4}\right]$	`[13,-16]`
8	$[-X, 0, -Y]$	$\left[\frac{1}{4}, \frac{1}{4}-x, \frac{1}{4}\right]$	`[14,-15]`
9	$[0, Y, -X]$	$\left[x+\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[17,-20]`
10	$[0, Y, X]$	$\left[\frac{3}{4}-x, \frac{3}{4}, \frac{1}{4}\right]$	`[18,-19]`
11	$[Y, X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}-x\right]$	`[21,-24]`
12	$[Y, -X, 0]$	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{3}{4}\right]$	`[22,-23]`

Table 20: Wyckoff bond: `48b@48f`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 0, 0]$	`[1,4]`
2	$[-X, 0, 0]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3]`
3	$[0, X, 0]$	$[0, x, 0]$	`[5,8]`
4	$[0, -X, 0]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	`[6,7]`
5	$[0, 0, X]$	$[0, 0, x]$	`[9,12]`
6	$[0, 0, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	`[10,11]`
7	$[0, X, 0]$	$\left[\frac{3}{4}, x+\frac{1}{4}, \frac{3}{4}\right]$	`[13,16]`
8	$[0, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}-x, \frac{1}{4}\right]$	`[14,15]`
9	$[X, 0, 0]$	$\left[x+\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[17,20]`
10	$[-X, 0, 0]$	$\left[\frac{3}{4}-x, \frac{3}{4}, \frac{1}{4}\right]$	`[18,19]`
11	$[0, 0, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}-x\right]$	`[21,24]`
12	$[0, 0, X]$	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{3}{4}\right]$	`[22,23]`

Table 21: Wyckoff bond: `96c@48f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, 0]$	`[1]`
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[3]`
4	$[X, -Y, -Z]$	$[x, 0, 0]$	`[4]`
5	$[Z, X, Y]$	$[0, x, 0]$	`[5]`

continued ...

Table 21

No.	vector	center	mapping
6	$[Z, -X, -Y]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[6]
7	$[-Z, -X, Y]$	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[7]
8	$[-Z, X, -Y]$	$[0, x, 0]$	[8]
9	$[Y, Z, X]$	$[0, 0, x]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[10]
11	$[Y, -Z, -X]$	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[11]
12	$[-Y, -Z, X]$	$[0, 0, x]$	[12]
13	$[Y, X, -Z]$	$\left[\frac{3}{4}, x+\frac{1}{4}, \frac{3}{4}\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{1}{4}, \frac{1}{4}-x, \frac{1}{4}\right]$	[14]
15	$[Y, -X, Z]$	$\left[\frac{1}{4}, \frac{1}{4}-x, \frac{1}{4}\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{3}{4}, x+\frac{1}{4}, \frac{3}{4}\right]$	[16]
17	$[X, Z, -Y]$	$\left[x+\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{3}{4}-x, \frac{3}{4}, \frac{1}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{3}{4}-x, \frac{3}{4}, \frac{1}{4}\right]$	[19]
20	$[X, -Z, Y]$	$\left[x+\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}-x\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{3}{4}\right]$	[22]
23	$[-Z, Y, X]$	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{3}{4}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}-x\right]$	[24]

* Wyckoff site: 48g, site symmetry: ..2

Table 22: Wyckoff bond: 48a@48g, bond symmetry: ..2

No.	vector	center	mapping
1	$[Y, X, X]$	$\left[\frac{1}{8}, y, \frac{1}{4}-y\right]$	[1,-19]
2	$[-Y, -X, X]$	$\left[\frac{7}{8}, \frac{1}{2}-y, \frac{3}{4}-y\right]$	[2,-17]
3	$[-Y, X, -X]$	$\left[\frac{3}{8}, y+\frac{1}{2}, y+\frac{3}{4}\right]$	[3,-20]
4	$[Y, -X, -X]$	$\left[\frac{5}{8}, -y, y+\frac{1}{4}\right]$	[4,-18]
5	$[X, Y, X]$	$\left[\frac{1}{4}-y, \frac{1}{8}, y\right]$	[5,-14]
6	$[X, -Y, -X]$	$\left[\frac{3}{4}-y, \frac{7}{8}, \frac{1}{2}-y\right]$	[6,-16]
7	$[-X, -Y, X]$	$\left[y+\frac{3}{4}, \frac{3}{8}, y+\frac{1}{2}\right]$	[7,-13]
8	$[-X, Y, -X]$	$\left[y+\frac{1}{4}, \frac{5}{8}, -y\right]$	[8,-15]
9	$[X, X, Y]$	$\left[y, \frac{1}{4}-y, \frac{1}{8}\right]$	[9,-24]
10	$[-X, X, -Y]$	$\left[\frac{1}{2}-y, \frac{3}{4}-y, \frac{7}{8}\right]$	[10,-22]
11	$[X, -X, -Y]$	$\left[y+\frac{1}{2}, y+\frac{3}{4}, \frac{3}{8}\right]$	[11,-23]
12	$[-X, -X, Y]$	$\left[-y, y+\frac{1}{4}, \frac{5}{8}\right]$	[12,-21]

Table 23: Wyckoff bond: 48b@48g, bond symmetry: ..2

No.	vector	center	mapping
1	$[0, X, -X]$	$\left[\frac{1}{8}, y, \frac{1}{4}-y\right]$	[1,19]
2	$[0, -X, -X]$	$\left[\frac{7}{8}, \frac{1}{2}-y, \frac{3}{4}-y\right]$	[2,17]
3	$[0, X, X]$	$\left[\frac{3}{8}, y+\frac{1}{2}, y+\frac{3}{4}\right]$	[3,20]
4	$[0, -X, X]$	$\left[\frac{5}{8}, -y, y+\frac{1}{4}\right]$	[4,18]

continued ...

Table 23

No.	vector	center	mapping
5	$[-X, 0, X]$	$\left[\frac{1}{4}-y, \frac{1}{8}, y\right]$	[5,14]
6	$[-X, 0, -X]$	$\left[\frac{3}{4}-y, \frac{7}{8}, \frac{1}{2}-y\right]$	[6,16]
7	$[X, 0, X]$	$\left[y+\frac{3}{4}, \frac{3}{8}, y+\frac{1}{2}\right]$	[7,13]
8	$[X, 0, -X]$	$\left[y+\frac{1}{4}, \frac{5}{8}, -y\right]$	[8,15]
9	$[X, -X, 0]$	$\left[y, \frac{1}{4}-y, \frac{1}{8}\right]$	[9,24]
10	$[-X, -X, 0]$	$\left[\frac{1}{2}-y, \frac{3}{4}-y, \frac{7}{8}\right]$	[10,22]
11	$[X, X, 0]$	$\left[y+\frac{1}{2}, y+\frac{3}{4}, \frac{3}{8}\right]$	[11,23]
12	$[-X, X, 0]$	$\left[-y, y+\frac{1}{4}, \frac{5}{8}\right]$	[12,21]

Table 24: Wyckoff bond: 96c@48g, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{8}, y, \frac{1}{4}-y\right]$	[1]
2	$[-X, -Y, Z]$	$\left[\frac{7}{8}, \frac{1}{2}-y, \frac{3}{4}-y\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{3}{8}, y+\frac{1}{2}, y+\frac{3}{4}\right]$	[3]
4	$[X, -Y, -Z]$	$\left[\frac{5}{8}, -y, y+\frac{1}{4}\right]$	[4]
5	$[Z, X, Y]$	$\left[\frac{1}{4}-y, \frac{1}{8}, y\right]$	[5]
6	$[Z, -X, -Y]$	$\left[\frac{3}{4}-y, \frac{7}{8}, \frac{1}{2}-y\right]$	[6]
7	$[-Z, -X, Y]$	$\left[y+\frac{3}{4}, \frac{3}{8}, y+\frac{1}{2}\right]$	[7]
8	$[-Z, X, -Y]$	$\left[y+\frac{1}{4}, \frac{5}{8}, -y\right]$	[8]
9	$[Y, Z, X]$	$\left[y, \frac{1}{4}-y, \frac{1}{8}\right]$	[9]
10	$[-Y, Z, -X]$	$\left[\frac{1}{2}-y, \frac{3}{4}-y, \frac{7}{8}\right]$	[10]
11	$[Y, -Z, -X]$	$\left[y+\frac{1}{2}, y+\frac{3}{4}, \frac{3}{8}\right]$	[11]
12	$[-Y, -Z, X]$	$\left[-y, y+\frac{1}{4}, \frac{5}{8}\right]$	[12]
13	$[Y, X, -Z]$	$\left[y+\frac{3}{4}, \frac{3}{8}, y+\frac{1}{2}\right]$	[13]
14	$[-Y, -X, -Z]$	$\left[\frac{1}{4}-y, \frac{1}{8}, y\right]$	[14]
15	$[Y, -X, Z]$	$\left[y+\frac{1}{4}, \frac{5}{8}, -y\right]$	[15]
16	$[-Y, X, Z]$	$\left[\frac{3}{4}-y, \frac{7}{8}, \frac{1}{2}-y\right]$	[16]
17	$[X, Z, -Y]$	$\left[\frac{7}{8}, \frac{1}{2}-y, \frac{3}{4}-y\right]$	[17]
18	$[-X, Z, Y]$	$\left[\frac{5}{8}, -y, y+\frac{1}{4}\right]$	[18]
19	$[-X, -Z, -Y]$	$\left[\frac{1}{8}, y, \frac{1}{4}-y\right]$	[19]
20	$[X, -Z, Y]$	$\left[\frac{3}{8}, y+\frac{1}{2}, y+\frac{3}{4}\right]$	[20]
21	$[Z, Y, -X]$	$\left[-y, y+\frac{1}{4}, \frac{5}{8}\right]$	[21]
22	$[Z, -Y, X]$	$\left[\frac{1}{2}-y, \frac{3}{4}-y, \frac{7}{8}\right]$	[22]
23	$[-Z, Y, X]$	$\left[y+\frac{1}{2}, y+\frac{3}{4}, \frac{3}{8}\right]$	[23]
24	$[-Z, -Y, -X]$	$\left[y, \frac{1}{4}-y, \frac{1}{8}\right]$	[24]

* Wyckoff site: 96h, site symmetry: 1

Table 25: Wyckoff bond: 96a@96h, bond symmetry: 1

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	[1]
2	$[-X, -Y, Z]$	$\left[-x, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[2]
3	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, y+\frac{1}{2}, -z\right]$	[3]
4	$[X, -Y, -Z]$	$\left[x+\frac{1}{2}, -y, \frac{1}{2}-z\right]$	[4]

continued ...

Table 25

No.	vector	center	mapping
5	$[Z,\, X,\, Y]$	$[z,\, x,\, y]$	[5]
6	$[Z,\, -X,\, -Y]$	$\left[z+\frac{1}{2},\, -x,\, \frac{1}{2}-y\right]$	[6]
7	$[-Z,\, -X,\, Y]$	$\left[-z,\, \frac{1}{2}-x,\, y+\frac{1}{2}\right]$	[7]
8	$[-Z,\, X,\, -Y]$	$\left[\frac{1}{2}-z,\, x+\frac{1}{2},\, -y\right]$	[8]
9	$[Y,\, Z,\, X]$	$[y,\, z,\, x]$	[9]
10	$[-Y,\, Z,\, -X]$	$\left[\frac{1}{2}-y,\, z+\frac{1}{2},\, -x\right]$	[10]
11	$[Y,\, -Z,\, -X]$	$\left[y+\frac{1}{2},\, -z,\, \frac{1}{2}-x\right]$	[11]
12	$[-Y,\, -Z,\, X]$	$\left[-y,\, \frac{1}{2}-z,\, x+\frac{1}{2}\right]$	[12]
13	$[Y,\, X,\, -Z]$	$\left[y+\frac{3}{4},\, x+\frac{1}{4},\, \frac{3}{4}-z\right]$	[13]
14	$[-Y,\, -X,\, -Z]$	$\left[\frac{1}{4}-y,\, \frac{1}{4}-x,\, \frac{1}{4}-z\right]$	[14]
15	$[Y,\, -X,\, Z]$	$\left[y+\frac{1}{4},\, \frac{3}{4}-x,\, z+\frac{3}{4}\right]$	[15]
16	$[-Y,\, X,\, Z]$	$\left[\frac{3}{4}-y,\, x+\frac{3}{4},\, z+\frac{1}{4}\right]$	[16]
17	$[X,\, Z,\, -Y]$	$\left[x+\frac{3}{4},\, z+\frac{1}{4},\, \frac{3}{4}-y\right]$	[17]
18	$[-X,\, Z,\, Y]$	$\left[\frac{3}{4}-x,\, z+\frac{3}{4},\, y+\frac{1}{4}\right]$	[18]
19	$[-X,\, -Z,\, -Y]$	$\left[\frac{1}{4}-x,\, \frac{1}{4}-z,\, \frac{1}{4}-y\right]$	[19]
20	$[X,\, -Z,\, Y]$	$\left[x+\frac{1}{4},\, \frac{3}{4}-z,\, y+\frac{3}{4}\right]$	[20]
21	$[Z,\, Y,\, -X]$	$\left[z+\frac{3}{4},\, y+\frac{1}{4},\, \frac{3}{4}-x\right]$	[21]
22	$[Z,\, -Y,\, X]$	$\left[z+\frac{1}{4},\, \frac{3}{4}-y,\, x+\frac{3}{4}\right]$	[22]
23	$[-Z,\, Y,\, X]$	$\left[\frac{3}{4}-z,\, y+\frac{3}{4},\, x+\frac{1}{4}\right]$	[23]
24	$[-Z,\, -Y,\, -X]$	$\left[\frac{1}{4}-z,\, \frac{1}{4}-y,\, \frac{1}{4}-x\right]$	[24]